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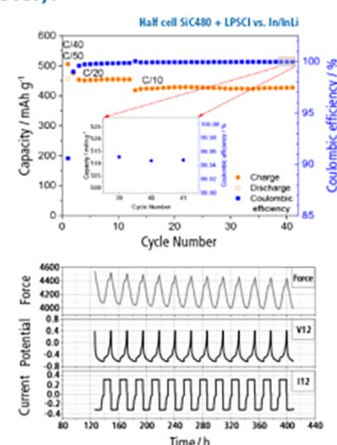
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Fire Detection and Localization Method Based on Deep Learning in Video Surveillance

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Abstract. Fire detection and localization in video surveillance had become a particularly important part of disaster rescue. Considering the fire detection and localization of slow detection speed, low detection accuracy, and low localization precision in video surveillance, we proposed a fire detection and localization method based on deep learning. The first thing we improved the SuperPoint method to extract video keyframe in video surveillance. The next thing we employed Convolutional Neural Network (CNN) model to detect the fire on the extracted video keyframes. The last thing we located the fire via superpixel and CNN on the extracted video keyframes which broke out a fire. The experimental results on open fire dataset revealed that the recall of keyframe extraction reached 0.83, the precision of fire detection reached 0.96 and the F1-score of fire localization reached 0.90. Our method realized rapid and accurate detection and precise localization of fire in video surveillance.

1. Introduction

Fire causes great harm to our life. With the quick growth of whole society, people start to care the role of disaster relief in urban safety management. As one of its core components, the importance of fire detection and localization in video surveillance is easy to see. Therefore, in order to deal with the fire disaster, reduce the disastrous consequences and explore effective response measures, it is very important to study fire detection and localization method in video surveillance.

With the rapid development of video surveillance and image processing technology, fire detection and localization via visual characteristics technology is also constantly move forward [1]. Through the intelligent surveillance system [2], fire can be detected and located accurately and in time, and relevant departments can be automatically notified [3]. Fire detection and localization via video images to detect hazard earlier and prevent fire better, which is extremely important in reality.

In recent decades, a great quantity methods of fire detection and localization had been proposed. In 1999, Yuan Hongyong et al. gave two technologies, namely "dual band image fire detection technology" and "light cross-sectional image smoke fire detection technology" [4]. The former realized earlier fire detection and true three-dimensional space localization. The latter realized the recognition and judgement of earlier fire. In 2006, Töreyn et al. proposed the space-time wavelet transform for discovering fire and non-fire areas [5]. A large number of heuristic thresholds were used in the method. In 2009, Ho proposed a smoke and fire detection method with shift detection algorithm



was carried out to discover possible fire [6]. In 2012, Yusuf Hakan et al. projected a method for fire detection via time and space message of image sequences [7]. The video was divided into space-time pieces, and then the covariance characteristics were extracted from the space-time pieces. The space-time characteristics of fire candidate areas were used as feature vectors, and fire detection was performed by support vector machines. In 2018, Li et al. projected a framework of fire detection via colour, motion, and twinkle of fires [8]. They employed Gaussian mixture model for fire detection. In 2019, Muhammad et al. proposed a primordial, economic, and highly efficient CNN model [9], which can perform fire detection, fire localization, and semantic comprehension of fire scenes.

In order to deal with the problems of slow detection speed, low detection accuracy or low localization precision in the above methods, we propose a fire detection and localization method based on deep learning in video surveillance. The main functions are organized as follows:

1. In order to avoid the tedious work of using traditional methods to detect and locate fire, we propose a framework of fire detection and localization based on deep learning, which exhibits good computational performance and processing power.
2. With the view of improving the speed of fire detection, we propose a keyframe extraction strategy. We improve the SuperPoint method to extract video keyframes and only perform fire detection and localization method on the extracted keyframes.
3. We perform fire detection via CNN, which improves the accuracy of fire detection, and we perform fire localization via superpixel and CNN, which improves the precision of fire localization.

The rest of this article is summarized as follows: in chapter 2, we propose a framework of fire detection and localization based on deep learning. Experimental results and discussion are obtained in chapter 3. Conclusions and future study are obtained in chapter 4.

2. Proposed framework

In this chapter, we describe the framework of fire detection and localization based on deep learning. Traditional fire detection and localization methods have slow detection speed, low detection accuracy and low localization precision, so we investigate deep learning models to solve these problems. The first thing we improve the SuperPoint method to extract video keyframe in video surveillance. The next thing we employ deep CNN model to detect the fire on the extracted video keyframes. The last thing we locate the fire via superpixel and CNN on the extracted video keyframes which broke out a fire. The framework of fire detection and localization is revealed in Figure 1.

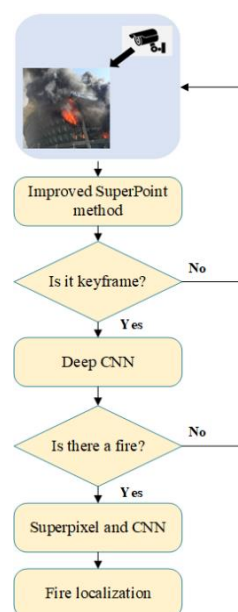


Figure 1. The framework of fire detection and localization.

2.1. Keyframe extraction

When extracting keyframe from video, the frame difference method and the clustering method are often used to achieve keyframe extraction. However, in video surveillance, traditional keyframe extraction methods often lose some important video frames or extract excessive video frames, which lead to poor performance of keyframe extraction. Therefore, we propose a keyframe extraction method via deep learning that improves the SuperPoint method to extract interest points and calculate descriptors, and extract keyframe according to the magnitude of the scene change in the video. Our method improves the performance of keyframe extraction and reduces the video keyframe redundancy.

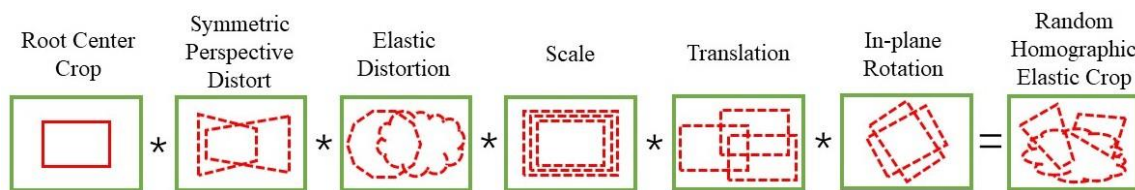
SuperPoint is a self-monitoring interest points detection and descriptors calculation method [10]. The interest points detected by network as the supervision information is used to train the network again. Homographic adaptation is the significant portion of SuperPoint method, which warp the input images to improve the performance of interest point detector. Homographic adaptation is divided into several sub transforms, including affine transform, scaling transform, translation transform and rotation transform, which together form homographic adaptation.

We employ elastic distortion to improve the SuperPoint method. The elastic distortion algorithm is first proposed by Patrice et al. in [11], which is applied to handwritten numeral recognition dataset. The elastic distortion generates random standard deviations for each dimensionality of the pixel, and employ gauss filter to filtrate the deviation matrix of each dimensionality, and finally use the amplification coefficient to control the deviation range. The formula expression of elastic distortion is as follows:

$$A'(x, y) = A(x, y) + \alpha(G_{\sigma} * (\Delta x, \Delta y)) \quad (1)$$

$A(x, y)$ shows the pixel first location. $A'(x, y)$ shows the pixel final location. Δx and Δy mean random displacement. $*$ means convolution operation. G_{σ} means a Gaussian of standard deviation σ . σ means the elasticity coefficient, and α means the amplification coefficient.

The homographic adaptation is suitable for rigid transformation, while the elastic distortion is suitable for non-rigid transformation which including flame change. Therefore, the elastic distortion has natural advantages for representing the flame change compared to the homographic adaptation. As a result, we make the elastic distortion to become a new sub transform, and form a new homographic adaptation together with other sub transforms, which improves the performance of interest point detector further and extracts truer video keyframe. The homographic adaptation after joining elastic distortion is revealed in Figure 2.

**Figure 2.** The homographic adaptation after joining elastic distortion.

With regard to the frame difference method, it is heavily dependent on the selected time interval. With regard to the clustering method, it is sensitive to noise and outliers. Hence the traditional keyframe extraction methods behave badly. SuperPoint is an interest points detection and descriptors calculation method, which can extract more accurate image local features via CNN. Therefore, we improve the SuperPoint method to extract video keyframes. The framework of keyframe extraction is revealed in Figure 3. The main procedures are as follows:

1. Define the current video frame A as keyframe and extract interest points and descriptors of frame A via the improved SuperPoint method.
2. Read the next video frame B and extract interest points and descriptors of frame B via the improved SuperPoint method.
3. Calculate the average distance D1 of corresponding interest points of frame A and frame B.
4. If the average distance D1 is greater than or equal to preset distance D2, make frame B as frame A and skip to step 1. If the average distance D1 is less than preset distance D2, skip to step 2. If the video reading is complete, it ends.

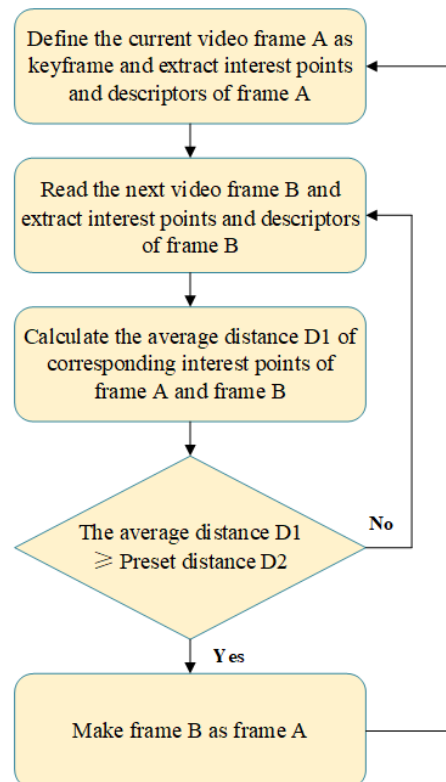


Figure 3. The framework of keyframe extraction.

2.2. Fire detection

Different from some earlier fire detection methods, we don't use the traditional method based on single feature or multi-feature fusion to detect fire, but use the deep learning method via CNN to detect fire. CNN has been used to handle a lot of video image problems and the performance is well. CNN is applied to detect fire, which can to a large extent improve the accuracy of fire detection. The CNN architecture for fire detection is revealed in Figure 4.

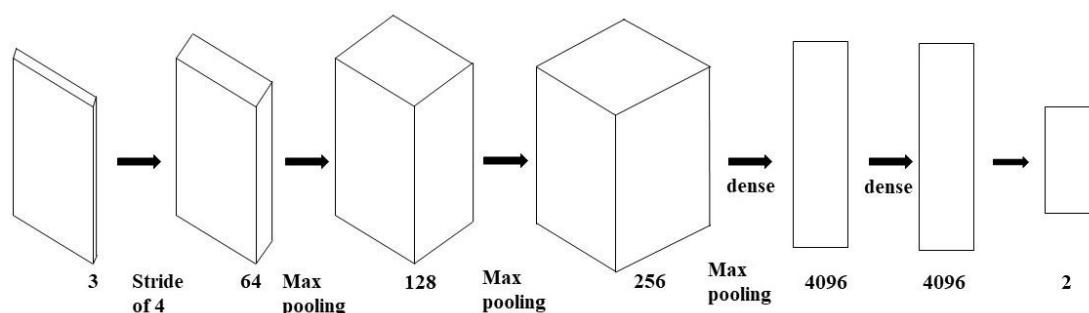


Figure 4. The CNN architecture for fire detection.

From Figure 4, the CNN architecture for fire detection has three convolutional layers, two full connection layers which have 4096 input connections, and a full connection layer which have 2 input connections. The size of three convolutional layers respectively is 64, 128, and 256, and max pooling layer is at the back of every convolutional layer.

The results of fire detection are revealed in Figure 5. Figure 5a shows the result of fire. If a fire is detected, there are fire word and red rectangular outline. Figure 5b shows the result of non-fire. If no fire is detected, there are non-fire word and green rectangular outline. The images in Figure 5 are selected from the Dunning's dataset [12].

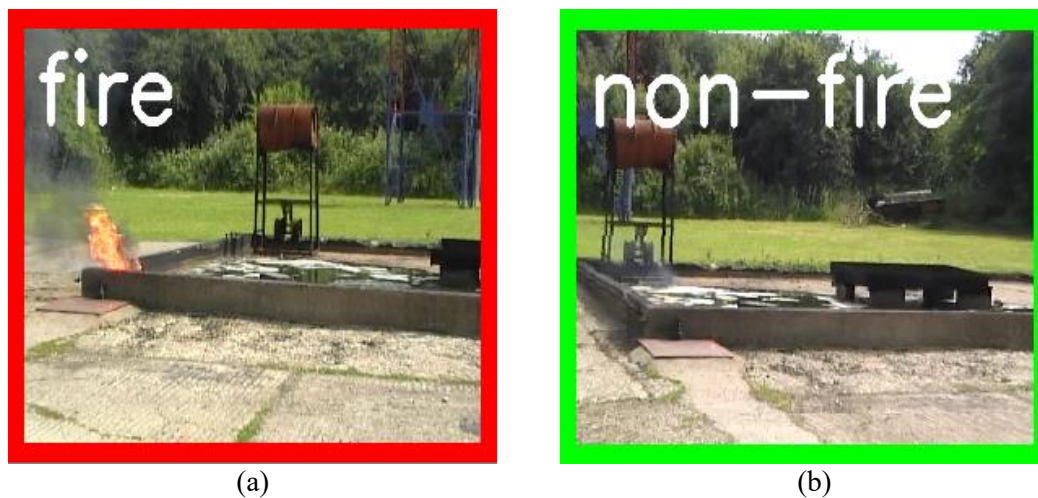


Figure 5. The results of fire detection. (a) The result of fire. (b) The result of non-fire.

2.3. Fire localization

The intuitive explanation of superpixel is to collect some pixels with similar features to generate a more typical large part, and novel part would turn into the fundamental unit of other further handling algorithms. The superpixel segmentation algorithm we use is the Simple Linear Iterative Clustering (SLIC) algorithm. The SLIC algorithm can create tight and equal superpixels, and has a high evaluation in view of computing speed, substance contour maintaining, and superpixel form, which can meet people's requirements. When keyframes are segmented into a lot of superpixel regions, we use CNN to detect the superpixel regions whether containing fire, and combine the superpixel regions which containing fire together to locate the fire. The CNN architecture for fire localization is revealed in Figure 6.

From Figure 6, the CNN architecture for fire localization has three consecutive management modules, and every separate module has the equal explanation. The identical unaltered collocation of pre-treatment and post-treatment layers are employed with three modules.

The results of fire localization are revealed in Figure 7. Figure 7a shows the segmentation result, and Figure 7b shows the localization result. The red regions mean non-fire and the green regions mean fire. From Figure 7, the SLIC algorithm segments keyframes into a lot of superpixels, and these superpixels realize fire localization via CNN. The images in Figure 7 are selected from the Dunning's dataset [12].

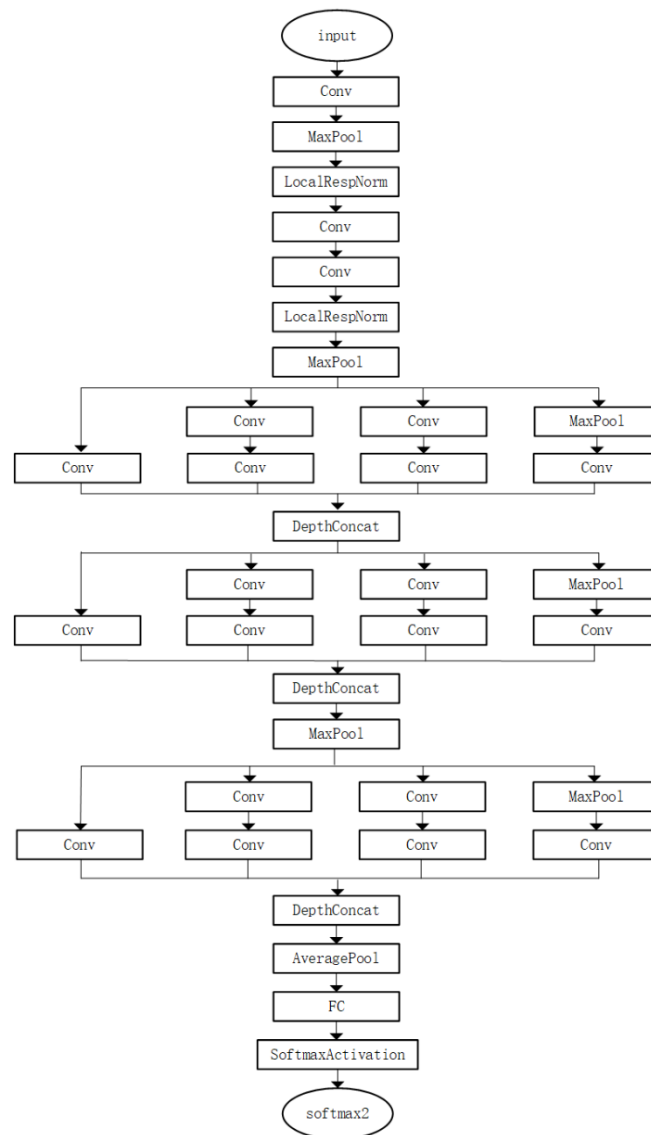


Figure 6. The CNN architecture for fire localization.

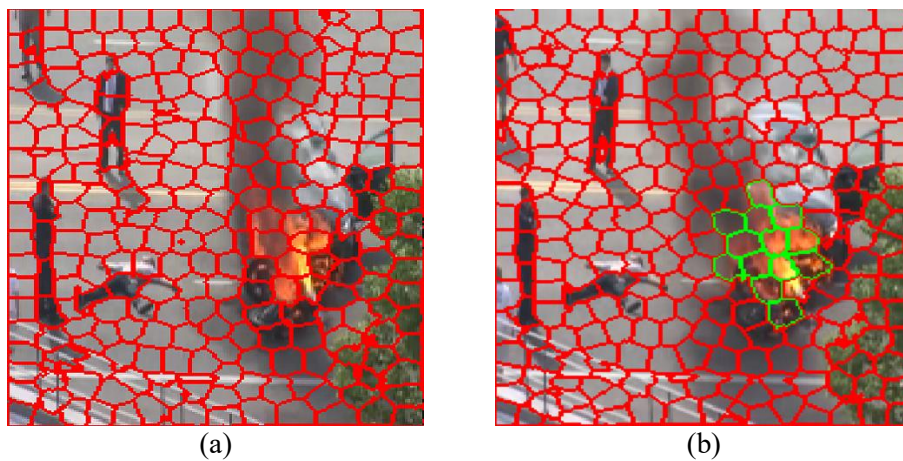


Figure 7. The results of fire localization. (a) The segmentation result. (b) The localization result.

3. Experimental results and discussion

3.1. Experimental setup

In view of proving the performance of our proposed method, we designed and conducted related experiments. We conduct the experiments with a system which has the configuration: NVidia GeForce GT 710 with 8 GB onboard memory and Windows 10 installed on an Intel Core i7 CPU with 16 GB RAM.

3.2. Experimental dataset

We select the Dunning dataset [12] as the experimental dataset, which includes 23408 images of fire and non-fire which are applied to training and test, 2931 images of fire and non-fire which are applied to validation, and 21 fire and non-fire videos. The images of experimental dataset are revealed in Figure 8. The first six images mean fires, and the last three images mean fire-like objects.



Figure 8. The images of experimental dataset.

3.3. Evaluation metrics

We choose True Positive Rate (TPR), False Positive Rate (FPR), Recall (R), Precision (P), Accuracy (A) and F1-score (F) to evaluate the performance. The formula expressions of evaluation metrics are as follows, where TP means true positive, TN means true negative, FP means false positive, and FN means false negative.

$$TPR = \frac{TP}{TP + FN} \quad (2)$$

$$FPR = \frac{FP}{FP + TN} \quad (3)$$

$$P = \frac{TP}{TP + FP} \quad (4)$$

$$R = \frac{TP}{TP + FN} \quad (5)$$

$$F = \frac{2 \cdot P \cdot R}{P + R} \quad (6)$$

$$A = \frac{TP + TN}{TP + FN + FP + TN} \quad (7)$$

3.4. Results and discussion

3.4.1. The performance of keyframe extraction methods. Firstly, we conduct experiment to verify the performance of the improved SuperPoint method to extract keyframe. We choose the frame difference method and the clustering method as comparison method. We choose recall, precision, and F1-score as evaluation metrics. The experimental results of different keyframe extraction methods are revealed in Table 1.

Table 1. The results of different keyframe extraction methods.

Method	R	P	F
Frame difference	0.42	0.50	0.46
Clustering	0.75	0.82	0.78
Improved SuperPoint	0.83	0.77	0.80

From Table 1, recall of the improved SuperPoint method reaches 0.83, which is higher than other methods. For F1-score, the improved SuperPoint method reaches 0.80, which also outperforms other methods. In terms of precision, the clustering method reaches 0.82, which is better than other methods, but our method also has certain competitiveness.

3.4.2. The performance of fire detection methods. Secondly, we conduct experiment to verify the performance of fire detection via CNN. We choose Celik et al. [13], Zhou et al. [14] and Chenebert et al. [15] proposed methods as comparison method. We choose true positive rate, false positive rate, F1-score, precision and accuracy as evaluation metrics. The experimental results of different fire detection methods are revealed in Table 2.

Table 2. The results of different fire detection methods.

Method	TPR	FPR	F	P	A
Celik et al. [13]	0.74	0.68	0.69	0.65	0.59
Zhou et al. [14]	0.99	0.98	0.78	0.64	0.64
Chenebert et al. [15]	0.99	0.28	0.92	0.86	0.89
Ours	0.92	0.26	0.94	0.96	0.90

From Table 2, for false positive rate, F1-score, precision and accuracy, our method all outperforms other methods, and respectively reach 0.26, 0.94, 0.96 and 0.90. For true positive rate, Zhou et al. and Chenebert et al. methods both reach 0.99, which are better than other methods, but our method also has certain competitiveness.

3.4.3. The performance of fire localization methods. Finally, we conduct experiment to verify the performance of fire localization via superpixel and CNN. We also choose Celik et al. [13], Zhou et al. [14] and Chenebert et al. [15] proposed methods as comparison method. We choose true positive rate, precision and F1-score as evaluation metrics. The experimental results of different fire localization methods are revealed in Table 3.

Table 3. The results of different fire localization methods.

Method	TPR	P	F
Celik et al. [13]	0.73	0.25	0.38
Zhou et al. [14]	0.44	0.02	0.04
Chenebert et al. [15]	0.98	0.83	0.90
Ours	0.90	0.89	0.90

From Table 3, precision of our method outperforms other methods, and reaches 0.89. For F1-score, Chenebert et al. and our methods both reach 0.90, which are better than other methods. In terms of true positive rate, Chenebert et al. method reaches 0.98, which is higher than other methods, but our method also has certain competitiveness.

On the whole, we propose the method that fire detection and localization based on deep learning achieves good performance that the recall of keyframe extraction reaches 0.83, the precision of fire detection reaches 0.96 and the F1-score of fire localization reaches 0.90. Therefore, our method can realize rapid and accurate detection and precise localization of fire in video surveillance.

4. Conclusion

Fire is very harmful to our life, as failing to control it can bring about giant disasters and result in human, economic and social losses. This is why we need to constantly study fire detection and localization methods. Therefore, we propose a fire detection and localization method based on deep learning. We conduct experiments on open fire dataset and the results show that keyframe extraction via improved SuperPoint method increases detection speed, fire detection via CNN increases detection accuracy and fire localization via superpixel and CNN increases localization precision.

We mainly study fire localization on video keyframe, and pay less attention to non-key video frame. In future studied, we will try to use tracking algorithm to locate fire of non-key video frame.

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